Computational Identification of High-Altitude Molecules Using Molecular Networking and Liquid Chromatography High-Resolution Mass Spectrometry (LC-HRMS): Developing Software by Chemists, for Chemists

Independent Study Thesis

Presented in Partial Fulfillment of the Requirements for
the Degree Bachelor of Arts in the
Departments of Chemistry & Computer Science at The College of
Wooster

by
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The College of Wooster
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WOOSTER

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Abstract

Include a short summary of your thesis, including any pertinent results. This section is *not* optional, and the reader should be able to learn the meat of your thesis by reading this (short) section.

This work is dedicated to the future generations of Wooster students.

Acknowledgments

No part of my journey at The College of Wooster has been accomplished alone. Behind every class, project, difficult conversation, and moment of growth are people who have supported me, challenged me, and reminded me why I started. This Independent Study is no exception. I am grateful to everyone who helped me become not only a better student and researcher, but also a better person.

First, I want to thank my Senior Independent Study advisors, Dr. Faust and Dr. Musgrave, for guiding me through a project that brings chemistry and computer science together. Dr. Faust, you have been one of my biggest champions. Thank you for believing in me, advocating for me, and helping me see what I am capable of even when I have struggled to see it myself. Dr. Musgrave, thank you for your guidance, patience, and willingness to work through the challenges of turning an interdisciplinary idea into a meaningful computer science project. I am also grateful to Dr. Paul Edmiston for bringing the original molecular-networking problem to my attention and giving me a reason to explore how the two disciplines I love could work together.

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To Fr. David, thank you for walking alongside me during my journey toward baptism. Your guidance helped me through a deeply personal part of my life, and I am grateful for the time, care, and faith you shared with me.

Most importantly, I thank my family and siblings, especially my seven sisters and six brothers. You are part of the reason I care so deeply about using what I learn to make a difference. Thank you for the love, sacrifices, patience, and encouragement that have carried me through every stage of my education. Whatever I accomplish, I will always carry my family with me.

Finally, to everyone whose name may not appear here but who has offered a kind word, a difficult lesson, a new opportunity, or a reason to keep going, thank you. This work may have my name on its cover, but the person who completed it was shaped by all of you.

Vita

Raelyn M. Brooks is a candidate for the Bachelor of Arts degree in Chemistry and Computer Science at The College of Wooster, with an anticipated graduation date of May 2027. His academic interests lie at the intersection of computational chemistry, software engineering, and scientific research. He is particularly interested in developing computational tools that allow scientists to investigate complex chemical problems more efficiently and reliably.

During his undergraduate studies, Brooks pursued interdisciplinary research combining chemistry and computer science. His Senior Independent Study, advised by Dr. Jennifer Faust and Dr. John Musgrave, focuses on developing molecular-networking software to support the investigation of unknown molecular features in remote environmental samples using liquid chromatography–high- resolution mass spectrometry (LC-HRMS). His previous research includes developing a Python-based cheminformatics pipeline using RDKit to explore multistep organic synthesis pathways across large chemical reaction datasets.

Brooks has gained professional experience in software engineering and cloud computing. As a Software Developer Intern at the Federal Reserve Bank of St. Louis, he contributed to production software development, automated infrastructure, and security remediation. He has also worked as an Associate Cloud Engineer at GoHub LLC, developing cloud infrastructure and software for a transportation platform. Additionally, he has served as a coding instructor at iD Tech, teaching programming, software development, and problem-solving to hundreds of students.

Outside the classroom, Brooks has been actively involved in campus leadership and community engagement. He has served as President of Beta Kappa Phi, participated in Black & Code, and contributed to student life through Residence Life and other campus organizations. These experiences have strengthened his commitment to mentorship, leadership, and creating opportunities for others.

Following graduation, Brooks intends to pursue a career in software engineering, particularly in developing computational tools for chemistry and scientific research. He also hopes to pursue graduate study in chemistry, computer science, or an interdisciplinary field connecting the two disciplines.

Fields of Study Major field: Chemistry & Computer Science

Specialization: Computational Chemistry, Molecular Networking, and Liquid Chromatography–High-Resolution Mass Spectrometry (LC-HRMS)

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Preface

I have always considered myself a problem solver. Whether I am writing code, working through a chemical reaction, or trying to understand why something does not work as expected, I am drawn to problems that require patience, creativity, and a willingness to approach things differently. This is ultimately what attracted me to both chemistry and computer science. Chemistry has taught me to think beyond what I can immediately observe and to consider the countless interactions that explain the world around us. Computer science has taught me that failure is often part of progress, and that solving a problem sometimes requires starting over, questioning an assumption, or finding an entirely different approach. Together, these disciplines have shaped how I think, how I learn, and the kind of work I hope to pursue.

The inspiration for this Independent Study came from a conversation with Dr. Paul Edmiston, who introduced me to a challenge involving molecular-networking software. His students were exploring how molecular networking could help identify chemically modified substances relevant to forensic toxicology and support investigations involving drug-facilitated crimes. However, understanding the computational processes behind the software and evaluating the reliability of its results presented a challenge. Dr. Edmiston needed someone willing to look beneath the surface of the software and understand how it worked. As a brother to seven sisters, the opportunity to contribute to research with potential implications for personal safety and justice resonated with me deeply. Beyond its scientific importance, I recognized a chance to apply my knowledge of chemistry and computer science to a problem that could make a meaningful difference in people's lives.

What made this opportunity especially exciting was that it brought together the two fields I had spent my undergraduate career studying. I have never viewed chemistry and computer science as competing interests. Instead, I see them as disciplines that strengthen one another. Chemistry presents complicated questions about the world, while computer science provides tools for investigating those questions at a scale and level of detail that would otherwise be difficult to achieve. Although the research eventually developed into a project centered on molecular networking and the identification of unknown features in remote environmental samples, that initial conversation remained central to my motivation: I wanted to understand how the software worked and how it could be made more useful and reliable for the scientists who depend on it.

My experiences outside the classroom have further shaped that goal. During my internship as a software developer at the Federal Reserve Bank of St. Louis, I learned the importance of writing dependable software, testing my work, and understanding the consequences of decisions made within complex systems. Working with cloud infrastructure and software architecture has shown me how careful design affects reliability and long-term usefulness. As a coding instructor, I have taught hundreds of students how to approach difficult programming problems, often reminding them of the same lesson I continue to learn myself: not understanding something immediately does not mean it cannot be understood. My previous computational chemistry work, including developing tools to explore multistep organic synthesis using large reaction datasets, also helped me recognize how powerful software can become when it is built with a clear understanding of the science it supports.

After graduation, I hope to pursue a career developing software for chemists and other scientific researchers. I want to build tools that do more than produce results. I want researchers to understand those results, explore the evidence behind them, and have confidence in the computational processes they use. This Independent Study represents my first substantial attempt to bring that ambition into practice. It has also challenged me to recognize that being both a chemist and a computer scientist does not mean having every answer. Instead, it means being willing to learn from both disciplines, recognize the limitations of each, and do the work necessary to connect them.

I would not have reached this point without the faculty members who have challenged, supported, and believed in me throughout my time at The College of Wooster.

Within the Computer Science Department, I want to thank Dr. Heather Guarnera for consistently helping me find ways to make what once seemed impossible feel achievable. Your encouragement has pushed me to think bigger about what I am capable of accomplishing. I thank Dr. Drew Guarnera, my first computer science professor, for helping establish the foundation upon which so much of my work now rests. I am also grateful to Dr. Palmer for contributing to my growth as a computer scientist and to Dr. Musgrave, my Senior Independent Study advisor, for guiding me through the computational challenges of this interdisciplinary project.

Within the Chemistry Department, I owe a special thank-you to Dr. Faust, my Senior Independent Study advisor and one of my greatest champions. Your belief in my abilities, your encouragement, and your willingness to help me grow have meant more than I can adequately express. I thank Dr. Martin, my academic advisor, for helping guide my academic journey and supporting my interests across two demanding disciplines. Dr. Snider, thank you for the difficult but necessary conversation that helped me recognize where I needed to grow. Sometimes the advice we need most is not the advice we expect to hear. I thank Dr. Bonvallet for being a lecturer whose teaching has consistently made learning chemistry engaging. I thank Dr. Gallo for

helping me fall in love with chemistry and recognize just how much there is to discover within it. Dr. Baker, thank you for being one of my favorite professors and for the simple kindness of always being someone I look forward to greeting. Finally, Dr. Feierabend, thank you for continuing to challenge me to pursue excellence, particularly during moments when I have struggled to recognize my own potential.

I also want to thank Dr. Edmiston for presenting the problem that ultimately helped me discover the direction I want to take my career. Sometimes the most important opportunities begin with a conversation and a question that nobody has fully answered.

This Independent Study is more than a requirement for graduation. It represents the intersection of my interests, my experiences, and the people who have helped shape my education. More importantly, it is a beginning. I entered college wanting to understand how things work. I hope to leave having taken my first meaningful steps toward building things that help others understand the world.

CHAPTER 1

Introduction

1.1 Research Problem and Motivation

Anthropogenic chemicals enter the environment through industrial activities, agricultural practices, and the manufacture and use of consumer products. These inputs include pharmaceuticals, pesticides, industrial chemicals, and plastic-associated compounds, among other substances. Some chemicals are sufficiently persistent or mobile to move beyond their immediate sources, complicating efforts to determine their distribution and environmental fate [6, 2, 10]. Their occurrence is not confined to urban or industrial settings. Investigations of mountain regions and remote waters demonstrate that chemical pollution can be detected far from major emission centers, although the relative contributions of long-range transport and local activities differ among sites [3, 11, 5].

The analytical problem extends beyond determining whether a sample contains selected contaminants. Environmental samples may produce many signals in liquid chromatography–high-resolution mass spectrometry (LC-HRMS), but detecting a signal does not establish the chemical identity of the substance responsible for it. Traditional targeted measurements are useful when the compounds of interest are known in advance; they are less suited to surveying chemical features that were not anticipated when a method was developed. Suspect screening and non-targeted analysis (NTA) broaden the range of features that can be investigated, yet incomplete reference libraries and uneven spectral coverage continue to constrain identification [6, 7, 4].

This limitation is especially important for remote environmental samples. A database match may support a candidate assignment for one feature, while nearby features in the same dataset remain unannotated despite containing potentially useful tandem mass spectrometry (MS/MS) information. Molecular networking offers a way to examine these data by representing features as nodes and retaining relationships between features with sufficiently similar MS/MS spectra. The resulting network can help researchers explore candidate chemical

relationships and prioritize features for further study, but a network connection is not itself a confirmed molecular identification [9, 1, 8].

The present Senior Independent Study addresses a second, connected problem: the need for a dependable computational workflow through which chemists can construct, inspect, preserve, and interpret molecular networks. For the College of Wooster's environmental HRMS research, useful software must connect the displayed network to the underlying spectral evidence and feature metadata while making its computational decisions reproducible. This project therefore combines environmental chemistry with software engineering. Its central contribution is the development and evaluation of a molecular-networking workflow and interface, using environmental HRMS data as both the motivating application and the basis for chemical case studies.

1.2 Environmental and Chemical Background

1.2.1 Long-Range Transport of Organic Contaminants

Organic contaminants can move away from their original sources through atmospheric processes. Their movement and deposition depend on properties such as volatility and persistence, as well as environmental conditions. Some semivolatile compounds can undergo repeated cycles of volatilization, atmospheric transport, and deposition, allowing them to reach regions distant from where they were emitted. Temperature gradients and precipitation can influence where these chemicals are deposited.

1.2.2 High-Altitude and Remote Surface-Water Environments

Remote, high-altitude environments provide useful settings for studying the distribution of environmental contaminants. Locations in the project's research context include Mount Everest, Mount Kilimanjaro, the Qinghai–Tibetan Plateau, and associated surface-water environments. Snow, precipitation, and surface water can provide evidence of atmospheric transport and deposition. Remoteness does not guarantee isolation from human-associated chemicals.

This project uses existing environmental HRMS data; sample collection is not itself a contribution of this thesis. The origins, sample types, and analytical details of the datasets used in the software evaluation will be documented in Chapter 2.

1.2.3 Previous Wooster Research and the Identification Gap

Previous research associated with the College of Wooster has investigated organic contaminants in remote and high-altitude environmental samples. This work provides the chemical and analytical context for the current

project. Even when LC-HRMS detects features of interest, not every feature can be assigned confidently to a specific compound. A feature may have accurate-mass, retention-time, isotope, or MS/MS information without a confirmed chemical identity.

The gap is both chemical and computational. Chemically, researchers need ways to prioritize and interpret features that lack confident identifications. Computationally, they need tools that allow spectral relationships and associated metadata to be examined consistently. This thesis addresses the computational part of that problem while using environmental data to demonstrate how the workflow can support chemical investigation.

1.3 LC-HRMS, Suspect Screening, and Non-Targeted Analysis

1.3.1 LC-HRMS and Tandem Mass Spectrometry

Liquid chromatography separates components of a sample before they enter a mass spectrometer. High-resolution mass spectrometry measures ions with high mass accuracy and provides information used to characterize molecular features. In tandem mass spectrometry (MS/MS), selected precursor ions are fragmented and the resulting product-ion spectra provide additional evidence about molecular structure. These data support chemical annotation and comparison, but their interpretation depends on data quality and available reference information.

1.3.2 Molecular Features Versus Identified Compounds

A molecular feature is an observed signal characterized by analytical attributes such as mass-to-charge ratio, retention time, and, when available, isotope or MS/MS information. A feature should not automatically be treated as a confirmed compound. Adduct formation, in-source fragmentation, isomers, incomplete spectra, and limited reference information can complicate interpretation.

This thesis uses the term *feature* for detected signals whose chemical identities have not been confirmed. The term *compound* is reserved for cases where the evidence supports that terminology. Network connections and candidate annotations are evidence to investigate, not substitutes for appropriate chemical confirmation.

1.3.3 Suspect Screening and Non-Targeted Analysis

Targeted analysis focuses on a predefined set of compounds. Suspect screening examines data for compounds on a suspect list, including compounds that may not be part of a conventional targeted method. Non-targeted analysis (NTA) aims to detect and investigate a broader range of chemical features without restricting the analysis to a fixed list of target compounds.

These approaches are complementary. Suspect screening can investigate compounds for which there is a prior reason to look, while NTA can retain information about features not anticipated in advance. Neither approach guarantees that every detected feature can be identified. Candidate assignments must be evaluated against available mass, retention-time, isotope, and fragmentation evidence, as applicable.

1.3.4 Limitations of Current Identification Approaches

Reference databases and spectral libraries are incomplete. A detected feature may not have a suitable reference spectrum, and different structures can sometimes produce similar or partially overlapping fragmentation patterns. Data quality, acquisition conditions, and processing choices can also affect the evidence available for interpretation.

Identifying and prioritizing unknown features therefore requires researchers to combine different types of evidence and preserve uncertainty in their conclusions. A computational tool can help organize data and make relationships easier to inspect, but it should not convert a similarity score or network connection into an unsupported claim of chemical identity.

1.4 Molecular Networking and Theoretical Foundations

1.4.1 MS/MS Fragmentation and Spectral Similarity

MS/MS spectra represent product ions generated when a precursor ion fragments under specified conditions. Related molecules may share some fragments or fragmentation patterns, so comparing spectra can provide clues about structural similarities. The strength of these inferences depends on spectrum quality, the comparison method, and the chemical context.

Spectral similarity is best treated as a way to prioritize and explore possible relationships. It is not, on its own, definitive evidence that two features represent the same molecule or lie on a specific transformation pathway.

1.4.2 Cosine Similarity and Molecular-Network Construction

A spectral similarity score quantifies the correspondence between two spectra under a specified comparison procedure. Cosine similarity is one common approach. After spectra are represented and peaks are matched according to defined rules, the score summarizes alignment between the resulting spectral vectors. The score depends on implementation choices such as peak matching, mass tolerances, preprocessing, and filtering.

In a molecular network, each node represents a molecular feature and an edge represents a relationship that meets the selected similarity criteria. Edge weights or labels can retain the similarity score so researchers can inspect the computational relationship. Thresholds affect the resulting graph: stricter criteria may remove weaker connections, while more permissive criteria may create a denser network. The chosen settings and their rationale should therefore be documented.

1.4.3 Graph Theory: Nodes, Edges, and Connected Components

Graph theory provides a formal way to represent relationships among objects. In molecular networking, nodes represent features and edges represent the spectral relationships retained by the workflow. A connected component is a set of nodes connected through paths in the graph. Isolated nodes have no retained edges under the selected network criteria.

These structures help summarize and navigate relationships in a dataset. However, a connected component should not automatically be interpreted as one chemical family, and an edge should not be treated as proof of shared identity or a reaction pathway. Interpretation requires considering the graph alongside spectral evidence, metadata, annotations, and analytical limitations.

1.4.4 Network Visualization and Interpretation

Network visualization can help researchers inspect complex relationships that may be difficult to explore in a table alone. Selecting nodes and edges, viewing feature metadata, examining similarity values, and filtering the network can connect the overall graph to the underlying chemical evidence. Visual encodings should communicate meaningful properties without implying more certainty than the data support.

The interface is most useful when users can move between the network-level view and the evidence behind a connection. Users should be able to inspect which features are connected, why an edge is present under the selected criteria, and what information is available about the features. Important settings and data provenance should be accessible enough to support repeatable analysis.

1.4.5 Limitations of Spectral Similarity as Evidence of Chemical Identity

A high spectral similarity score indicates agreement under the selected computational comparison; it does not independently establish molecular identity. Similar spectra can arise for different structures, and structurally related compounds can produce different spectra under different conditions. Missing or low-quality MS/MS data may also weaken or prevent meaningful comparisons.

The software developed in this project is intended to support exploration of evidence rather than replace

expert chemical judgment or confirmatory experiments. Where reference standards or other appropriate confirmation are absent, the thesis will distinguish tentative annotations and possible relationships from confirmed identifications.

1.5 Existing Software and Research Gap

1.5.1 MS-DIAL and GNPS in the Existing Workflow

The existing research workflow uses MS-DIAL for upstream processing of mass-spectrometry data and molecular-networking tools such as the Global Natural Products Social Molecular Networking platform (GNPS) to explore spectral relationships. These tools address different stages of the workflow: processing software prepares and organizes analytical data, while molecular-networking software represents relationships among features.

This project builds on that workflow rather than assuming that any one tool is universally unsuitable. The relevant question is whether the workflow used for the College of Wooster's environmental HRMS research meets practical requirements for data handling, network inspection, reproducibility, and interpretation.

1.5.2 Documented Limitations in the Wooster Research Workflow

The motivation for this project includes workflow difficulties reported during local molecular-networking work, particularly around preserving work and reliably saving or exporting results. These concerns should be described as observed or reported issues in this specific research context, not as a general claim that GNPS or other existing platforms fail for all users.

The project will translate relevant research needs into explicit software requirements. These may include dependable input handling, inspectable node and edge information, links between network elements and chemical metadata, and clear behavior when saving or exporting results. The final requirements should reflect documented needs and features actually implemented and evaluated.

1.5.3 The Need for Software Designed Around Chemists' Needs

Molecular-networking software must do more than generate a graph. It should help users move between a network-level view and the underlying evidence, understand how connections were calculated, and preserve the information needed to reproduce an analysis. The interface should make important details accessible without requiring users to reconstruct the workflow from separate files or undocumented steps.

This project treats software reliability and usability as research concerns alongside the chemical application. The proposed contribution is a workflow and interface tailored to the current environmental HRMS research context, evaluated through computational tests and case studies using real data. Its success will be assessed based on implemented functionality and measured results rather than assumed in advance.

1.6 Research Question and Specific Aims

The central research question is:

> What computational workflow and interface are needed to make molecular networking reliable, reproducible, and useful for Wooster's environmental HRMS research?

The project addresses this question through four aims:

1. **Define the chemistry and data requirements.** Identify the analytical data, metadata, and evidence needed to construct and interpret molecular networks for the environmental HRMS research context.
2. **Develop an end-to-end molecular-networking workflow.** Implement a workflow that processes supported input data, compares MS/MS spectra, constructs a network, and allows users to inspect network elements and associated chemical information.
3. **Evaluate computational correctness, reproducibility, performance, and workflow usefulness.** Test calculations, input handling, graph construction, repeatability, performance, and relevant user-facing tasks using documented evaluation methods.
4. **Apply the software to environmental HRMS case studies.** Use selected environmental data to examine annotated and unidentified features, describe relationships supported by the evidence, and identify remaining uncertainties.

1.7 Project Scope and Thesis Contributions

1.7.1 Project Scope

This project focuses on computational support for molecular networking using existing environmental HRMS data. It includes defining data requirements, implementing and testing the network workflow, designing an interface for inspecting features and spectral relationships, and applying the software to selected chemical case studies. The specific input formats, data-processing assumptions, software features, and evaluation procedures will be defined in Chapter 2.

The project does not assume that a network alone can confirm the identity of unknown compounds, prove chemical transformation pathways, or replace reference standards and other confirmatory methods. Software development alone also does not establish the environmental source, toxicity, or ecological effect of a detected feature. Claims in the chemical case studies will be limited to what the available analytical data and supporting evidence justify.

1.7.2 Expected Computer Science Contributions

The Computer Science contribution is the design, implementation, and evaluation of a molecular-networking workflow and interface for the stated research context. The contribution will be documented through the software architecture, similarity and graph-construction methods, interface behavior, input validation, testing, and reproducibility practices. Chapter 3 will report what was implemented and how it performed under the evaluation procedures.

1.7.3 Expected Chemistry Contributions

The Chemistry contribution is the application of the workflow to environmental HRMS data to investigate spectral relationships among annotated and unidentified molecular features. The case studies will describe the evidence available for selected features, explain how network relationships may help prioritize further investigation, and state the remaining limitations. Tentative network relationships will not be presented as confirmed chemical identifications.

1.7.4 Organization of the Thesis

Chapter 2 describes the research design, environmental datasets and analytical methods, upstream processing, software requirements and implementation, network construction, and evaluation procedures. Chapter 3 presents the dataset summary, software implementation results, computational validation, performance and workflow evaluation, network characteristics, and chemical case studies. Chapter 4 interprets the computational and chemical findings, discusses design trade-offs and limitations, and considers implications for future environmental HRMS research. Chapter 5 summarizes the contributions, answers the central research question, and identifies future directions.

CHAPTER 2

Methodology, Research Design, and Software Development

2.1 Overall Research Design and Workflow

2.2 Environmental Samples and Chemistry Experimental Methods

2.3 Upstream Data Processing and Chemical Annotation

2.4 Software Requirements and System Architecture

2.5 Spectral Comparison and Similarity Calculation

2.6 Molecular-Network Construction

2.7 User Interface Design and Interaction

2.8 Software Implementation and Reproducibility

2.9 Software Evaluation and Testing Methods

2.10 Chemical Case-Study Design

CHAPTER 3

Results and Data Analysis

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3.2 Software Implementation Results

3.3 Computational Correctness and Validation

3.4 Software Performance and Workflow Evaluation

3.5 Molecular-Network Structure and Characteristics

3.6 Chemical Case Study 1: Annotated Molecular Features

3.7 Chemical Case Study 2: Unidentified Molecular Features

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4.2 Computational Contributions and Design Trade-Offs

4.3 Chemical Significance and Comparison with Literature

4.4 Study and Implementation Limitations

4.5 Implications for Future Environmental HRMS Research

CHAPTER 5

Conclusions and Future Work

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5.2 Principal Computer Science Contributions

5.3 Principal Chemistry Contributions

5.4 Answer to the Central Research Question

5.5 Future Work

5.6 Final Remarks

References

[1] Wout Bittremieux et al. "Comparison of Cosine, Modified Cosine, and Neutral Loss Based Spectrum Alignment For Discovery of Structurally Related Molecules". In: *Journal of the American Society for Mass Spectrometry* 33.9 (2022), pp. 1733–1744. DOI: 10.1021/jasms.2c00153. URL: https://doi.org/10.1021/jasms.2c00153 (page 2).

[2] "Environmental Contaminants: Using Natural Archives to Track Sources and Long-Term Trends of Pollution". In: (2015). Ed. by Jules M. Blais, Michael R. Rosen, and John P. Smol. DOI: 10.1007/978-94-017-9541-8 (page 1).

[3] Gillian L. Daly and Frank Wania. "Organic Contaminants in Mountains". In: *Environmental Science & Technology* 39.2 (2005), pp. 385–398. DOI: 10.1021/es048859u (page 1).

[4] Henriette Hupatz, Ingrid Rahu, Wen-Chao Wang, et al. "Critical review on in silico methods for structural annotation of chemicals detected with LC/HRMS non-targeted screening". In: *Analytical and Bioanalytical Chemistry* 417.2 (Jan. 2025), pp. 473–493. DOI: 10.1007/s00216-024-05471-x. URL: https://doi.org/10.1007/s00216-024-05471-x (page 1).

[5] Zhiyuan Ma et al. "Human activity driven spatial heterogeneity of pollutants in a highland river basin revealed by non-target screening and improved PBT evaluation: A case study from the Yalong River on the Qinghai-Tibet Plateau, China". In: *Journal of Hazardous Materials* 496 (2025), p. 139493. ISSN: 0304-3894. DOI: 10.1016/j.jhazmat.2025.139493. URL: https://www.sciencedirect.com/science/article/pii/S0304389425024094 (page 1).

[6] K. E. Manz et al. "Non-targeted Analysis (NTA) and Suspect Screening Analysis (SSA): A Review of Examining the Chemical Exposome". In: *Journal of Exposure Science & Environmental Epidemiology* 33.4 (2023), pp. 524–536. DOI: 10.1038/s41370-023-00574-6. URL: https://www.nature.com/articles/s41370-023-00574-6 (page 1).

[7] Bastian Schulze et al. "An assessment of quality assurance/quality control efforts in high resolution mass spectrometry non-target workflows for analysis of environmental samples". In: *TrAC Trends in Analytical Chemistry* 133 (2020), p. 116063. ISSN: 0165-9936. DOI: https://doi.org/10.1016/j.trac.

2020.116063. URL: https://www.sciencedirect.com/science/article/pii/S0165993620302922 (page 1).

[8] Emma L. Schymanski et al. "Identifying Small Molecules via High Resolution Mass Spectrometry: Communicating Confidence". In: *Environmental Science & Technology* 48.4 (2014), pp. 2097–2098. DOI: 10.1021/es5002105. URL: https://doi.org/10.1021/es5002105 (page 2).

[9] Mingxun Wang et al. "Sharing and Community Curation of Mass Spectrometry Data with Global Natural Products Social Molecular Networking". In: *Nature Biotechnology* 34.8 (2016), pp. 828–837. DOI: 10.1038/nbt.3597 (page 2).

[10] John L. Wilkinson et al. "Pharmaceutical pollution of the world's rivers". In: *Proceedings of the National Academy of Sciences* 119.8 (2022), e2113947119. DOI: 10.1073/pnas.2113947119. eprint: https://www.pnas.org/doi/pdf/10.1073/pnas.2113947119. URL: https://www.pnas.org/doi/abs/10.1073/pnas.2113947119 (page 1).

[11] Yue Xu et al. "Long-range Atmospheric Transport of Persistent Organochlorinated Compounds to a Remote Mountain Site in Southwestern China". In: *Journal of Environmental Monitoring* 13 (2011), pp. 3119–3127. DOI: 10.1039/c1em10470e (page 1).